

A Benchmark for Planetary Atmospheric Dynamics: New Constraints on Titan's Atmospheric Seasonal Variability

Abstract:

Titan's atmosphere exhibits seasonal variations due to changes in its circulation. These variations impact Titan's methane-based hydrocarbon cycle, organic haze distribution, and atmospheric temperature structure. Previously, the Cassini and JWST conducted detailed observations of Titan's atmosphere during northern winter, spring, and summer. However, Titan's atmosphere was scarcely observed during northern fall, preventing a complete understanding of Titan's seasonal cycles. We propose a JWST/FGS stellar occultation campaign for December 12, 2027, which will provide robust constraints on Titan's atmosphere during northern fall. Using precision astrometry, FGS will capture high-resolution images within the 1-5 μm spectral range, encompassing absorption and emission bands from a multitude of species in Titan's atmosphere. We will develop a sophisticated radiative transfer model to retrieve atmospheric number density and temperature profiles for these species. By exploiting a rare, transient stellar alignment, this program will analyze Titan's atmosphere at higher vertical resolution than other techniques. This program's results will advance models of Titan's climate and photochemistry throughout Titan's entire seasonal cycle. The similarities between Titan's atmosphere and those of other planets, particularly early Earth's atmosphere, enable the results of this program to broadly inform the seasonal evolution of Titan-like atmospheres within and beyond the solar system.

- **Scientific Justification**

Titan's Atmosphere Experiences Significant Seasonal Variations

Cassini observations revealed that Titan's atmosphere experiences significant seasonal variations over cycles of 29.45 Earth years. Titan's lower atmosphere (100-600 km altitude) was observed by the Composite Infrared Spectrometer (CIRS) instrument. CIRS measurements demonstrated the coexistence of two circulation cells upwelling at the midlatitudes and downwelling at both poles from January 2010 to June 2010 (Vinatier et. al., 2015). Two years later, global circulation totally reversed, resulting in a single pole-to-pole upwelling at the north pole and downwelling at the south pole. During Titan's southern autumn in 2015, CIRS observations indicated a compositional enhancement in photochemical compounds in the stratosphere and mesosphere above the south pole (Mathé et. al., 2020). This trend was accompanied by a continuous decrease in photochemical compound abundances above Titan's northern hemisphere until the end of the Cassini mission in 2017 (Coustenis et. al., 2020). Seasonal temperature changes were maximal at high latitudes (Sylvestre et. al., 2020). For example, temperatures at 70°S decreased by 19 K between northern summer (2007) and late autumn (2016) at the 15 mbar pressure-level altitude. Because these seasonal variations impact the composition and energy balance in Titan's atmosphere, they influence the evolution of Titan's methane-based hydrocarbon cycle, organic haze distribution, and atmospheric escape.

Although Cassini and JWST extensively observed Titan's atmosphere during northern winter, spring, and summer (Lorenz et. al., 2008; Nixon et. al., 2025), detailed observations of Titan's atmosphere during northern fall are currently unavailable. Currently, Titan is in northern fall, and our observing capabilities have significantly improved since Titan's previous northern fall equinox. For the first time, it is possible to complete detailed observations of Titan during all four of its seasons.

We propose a JWST Fine Guidance Sensor (FGS) Titan stellar occultation campaign for December 12, 2027. Although this occultation should be visible to ground-based observatories around the Indian Ocean (French and Souami, 2023), JWST will not be hindered by light pollution, atmospheric turbulence, or inclement weather. JWST will also provide higher spectral and spatial resolution than HST or ground-based facilities. By providing precision astrometry, FGS will be the only instrument capable of observing the stellar occultation. This project will aim to retrieve well-constrained temperature and composition profiles for Titan's lower and middle atmosphere during northern fall. These profiles can be incorporated into future models of Titan's climate and photochemistry, as well as models of Titan-like exoplanet atmospheres. Because Titan's atmosphere is similar to other planetary atmospheres, particularly early Earth's atmosphere, results from this program can contextualize the seasonal evolution of planetary atmospheres.

Lack of Detailed Observations During Titan's Northern Fall

Prior Cassini, JWST, and ground-based measurements provided robust constraints on Titan's atmosphere during northern winter, spring, and summer. However, Titan's atmosphere was scarcely observed during northern fall. Following Cassini's Grand Finale observations of Saturn and its rings, the mission ended in September 2017 (Ip et. al., 2025). The most recent

published HST Titan observations occurred between 2017-2019 (Karkoschka, 2022), while the most recent published JWST Titan observations occurred between 2022-2023 (Nixon et. al., 2025). These events all occurred within Titan's northern summer. Periodic, detailed observations of Titan's atmosphere are critical for improving global circulation models (GCMs) and photochemical models of Titan's atmosphere, which can help to characterize atmospheric circulation and organic haze production.

Currently, the planetary science community lacks observations of Titan's atmosphere with similar levels of detail as Cassini or JWST. These observations are necessary for filling gaps in our understanding of Titan's seasonal cycle. During Titan's previous northern fall (November 1995 – October 2002), no in-situ measurements occurred, and our observing capabilities have greatly improved since then. Titan is currently in northern fall, presenting an opportunity to conduct detailed observations of Titan during a previously remotely investigated phase of Titan's orbit. Therefore, this program will help to understand the evolution of Titan's atmosphere throughout its entire seasonal cycle.

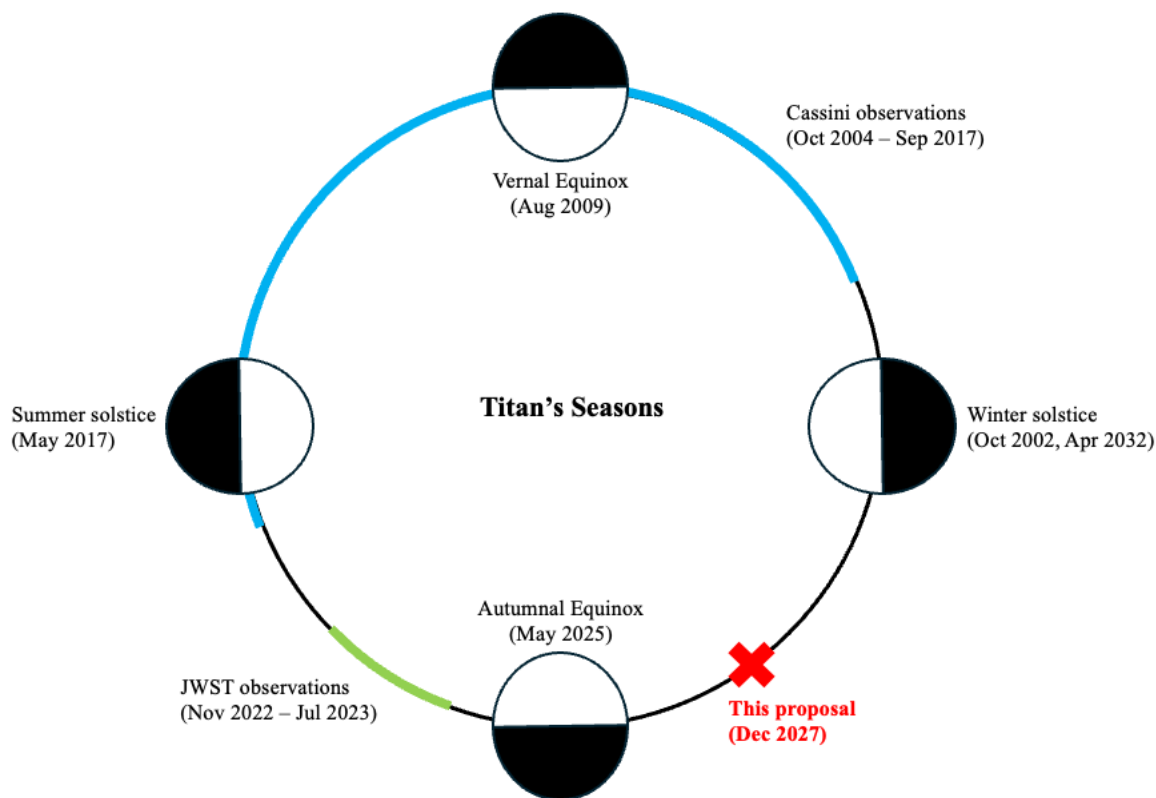


Figure 1: The proposed JWST measurements will occur during the final phase of Titan's orbit lacking in-depth observations, facilitating detailed analyses of Titan's seasonally evolving atmosphere. The four circles represent Titan at the beginning of each season, and the light and dark areas represent Titan's dayside and nightside. The blue arc represents Cassini-Huygens observations that occurred during northern winter and spring (figure adapted from Lorenz et. al., 2010), while the green arc represents prior JWST observations during northern summer. In contrast, Titan's atmosphere has been sparsely studied during northern fall. This program will robustly constrain Titan's atmospheric and

temperature profiles through JWST/FGS observations of the December 12, 2027 stellar occultation (indicated by the red “X” in the figure), during Titan’s northern fall. These observations are critical to enable accurate models of Titan’s climate and photochemistry throughout Titan’s entire seasonal cycle.

This Proposal: JWST/FGS Titan Stellar Occultation Campaign

We propose JWST Fine Guidance Sensor (FGS) observations of an upcoming Titan stellar occultation, predicted to occur on December 12, 2027. This project will target Titan’s lower and middle atmosphere, a region characterized by the absorption of solar infrared radiation by the organic haze, during northern fall. During the event, FGS will track a background star (French and Souami, 2023), identified in the Gaia Data Release 3 (Gaia Collaboration, 2023) catalogue (ID: 2564400353966267904). FGS will capture 2.3’ x 2.3’ images within the 1-5 μm wavelength range as the star passes behind both sides of Titan’s limb. Using these images, we will obtain spectra from different tangent altitudes in Titan’s atmosphere. Our tangent altitude grid will range between 0-600 km, with ~ 10 km vertical resolution. Using the NEMESIS radiative transfer code (Irwin et. al., 2008), we will retrieve N_2 , CH_4 , HCN , C_2H_2 , and CO density profiles from the spectra; NEMESIS has been applied to prior Titan atmospheric retrievals (Nixon et. al., 2025). Since Titan’s atmosphere is dominated by N_2 , we will utilize the N_2 density profiles to derive temperature profiles, assuming hydrostatic equilibrium.

The unique capabilities of JWST and FGS will enable us to conduct detailed atmospheric retrievals during Titan’s northern fall. Compared to other space-based facilities, JWST has the highest spectral and spatial resolution. By providing precision astrometry, FGS will be the only JWST instrument capable of observing the occultation. Although the occultation is expected to be visible to ground-based facilities near the Indian Ocean (French and Souami, 2023), their observations could be plagued by atmospheric turbulence, high cloud cover, or light pollution. In contrast, JWST will not suffer from these problems.

This project will obtain well-constrained density and temperature profiles in Titan’s lower and middle atmosphere with a vertical resolution of ~ 10 km. Furthermore, we will obtain evenly-spaced spectra on both sides of Titan’s limb, allowing future reanalysis. As a result, this project will provide the most robust constraints on Titan’s northern fall atmosphere to date. By comparing these results with those inferred from prior retrievals, we will create the most detailed, global picture of Titan’s seasonally evolving atmosphere to date.

- **Broader Impacts and Implications**

New Constraints on Titan’s Seasonal Variability: Implications for Planetary Atmospheres

This program will acquire new density and temperature profiles that will contextualize seasonal variability in Titan’s atmosphere. These profiles can be incorporated into future Titan global circulation and photochemical models and provide constraints on the sources and sinks of energy within Titan’s atmosphere. In addition, the profiles will provide a benchmark for Titan’s lower and middle atmosphere prior to the arrival of Dragonfly.

Titan is currently in northern fall, presenting an opportunity to conduct detailed Titan observations during a poorly observed phase of its orbit. In addition to its timeliness, this proposal directly addresses the key science goals of JWST’s “Other Worlds” theme: “Measure the physical and chemical properties of planetary systems, including our own solar system, and investigate the potential for life in those systems”. Within Titan’s atmosphere, aerosols within photochemically-generated organic haze descend to Titan’s surface, potentially engaging in prebiotic chemistry with materials from that environment (Raulin et. al., 2012). Because the species in our density retrieval are precursors to the organic haze, this project will constrain the availability of biologically relevant molecules. Because Titan’s atmosphere is a benchmark for “exo-Titans” (Ranjan et. al., 2025), this project will offer insights into an exoplanet population that has only recently been explored (Turbet et. al., 2018; Mandt et. al., 2022; Thompson et. al., 2022). Furthermore, since early Earth’s atmosphere and Titan’s atmosphere have similar compositions, these observations will inform our understanding of early Earth’s atmosphere.

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